

Bin Liu

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BASIC INFORMATION

Date of Birth	April 1990
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EDUCATION

- 2009.09-2013.06 B.S. in Statistics, Shandong University.
- 2013.09-2019.06 Ph.D. in Probability Theory and Mathematical Statistics, School of Management, Fudan University.
- 2017.09-2018.09 Visiting Scholar, Department of Statistics and Operations Research, University of North Carolina at Chapel Hill, USA.

EMPLOYMENT

- 2019.07-2020.07 Postdoctoral Fellow, Department of Statistics, The Chinese University of Hong Kong.
- 2020.07-2023.07 Lecturer, Department of Statistics and Data Science, School of Management, Fudan University.
- 2023.12-Present Associate Professor, Department of Statistics and Data Science, School of Management, Fudan University.

TEACHING

- Undergraduate courses: Statistical Software; The Path to a Strong Nation: Situation, Policy, and Mission III; The Path to a Strong Nation: Situation, Policy, and Mission IV.
- Master of Applied Statistics courses: Data Mining and Machine Learning; Statistical Computing; Business Practice in Data Science.
- MPAcc course: Business Forecasting.

RESEARCH INTERESTS

High-dimensional statistical inference; change-point testing and inference; data-adaptive testing; robust methods; machine learning; diffusion processes; reinforcement learning.

PUBLICATIONS

Published / Accepted * denotes corresponding author

- (1) Bin Liu; Zhengling Qi; Xinsheng Zhang; Yufeng Liu*. Change point detection for high-dimensional linear models: a general tail-adaptive approach. *Statistica Sinica*, 2026, 36(4): 1-40.
- (2) Bin Liu; Xinsheng Zhang; Yufeng Liu*. Simultaneous change point detection and identification for high-dimensional linear models. *Statistica Sinica*, 2026, 36(2): 1-40.
- (3) Xianru Wang; Bin Liu*; Xinsheng Zhang; Yufeng Liu. Efficient multiple change point detection and localization for high-dimensional quantile regression with heteroscedasticity. *Journal of the American Statistical Association*, 2025, 120(550): 976-989.
- (4) Zhi Yang; Liwen Zhang; Siyu Sun; Bin Liu*. Robust change point detection for high-dimensional linear models with tolerance for heavy tails. *The Canadian Journal of Statistics*, 2025, 53(1): 1-20.
- (5) Haixu Ma; Zhuoyu Shi; Minjeong Kim; Bin Liu; Patrick J. Smith; Yufeng Liu; Guorong Wu. Disentangling sex-dependent effects of APOE on diverse trajectories of cognitive decline in Alzheimer's disease. *NeuroImage*, 2024, 292: 120609.
- (6) Xianru Wang; Bin Liu*; Xinsheng Zhang; Yufeng Liu. Efficient change point detection for high-dimensional generalized linear models. *The Canadian Journal of Statistics*, 2023, 51(2): 569-629.
- (7) Xianru Wang; Bin Liu*; Xinsheng Zhang. A computationally efficient and flexible algorithm for high-dimensional mean and covariance matrix change point models. *Journal of the Korean Statistical Society*, 2022, 51: 1216-1246.
- (8) Bin Liu; Xinsheng Zhang; Yufeng Liu*. High-dimensional change point inference: recent developments and extensions. *Journal of Multivariate Analysis*, 2022, 188: 1-20.
- (9) Bin Liu; Xinsheng Zhang; Yufeng Liu*. Simultaneous change point inference and structure recovery for high-dimensional Gaussian graphical models. *Journal of Machine Learning Research*, 2021, 22: 1-62.
- (10) Bin Liu; Cheng Zhou; Xinsheng Zhang; Yufeng Liu*. A unified data-adaptive framework for high-dimensional change point detection. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 2020, 82(4): 933-963.
- (11) Bin Liu; Cheng Zhou*; Xinsheng Zhang. A tail-adaptive approach for change point detection. *Journal of Multivariate Analysis*, 2019, 169: 33-48.
- (12) Mingjuan Zhang; Cheng Zhou; Yong He; Bin Liu. Data-adaptive test for high-dimensional multivariate analysis of variance problem. *Australian & New Zealand Journal of Statistics*, 2018, 60: 447-470.
- (13) Yong He; Xinsheng Zhang; Jiadong Ji; Bin Liu*. Joint estimation of multiple high-dimensional Gaussian copula graphical models. *Australian & New Zealand Journal of Statistics*, 2017, 59: 289-310.

Submitted / Under Revision

- (1) Bin Liu; Yufeng Liu*. A general U-statistic framework for high-dimensional multiple change-point. Under revision at *Biometrics*.
- (2) Xianru Wang; Bin Liu*; Xinsheng Zhang; Yufeng Liu. Efficient oracle-based estimation and inference for multiple change points in high-dimensional Gaussian graphical models under temporal dependence. Under revision at *Statistica Sinica*.

- (3) Bin Liu*; Yufeng Liu; Zhiqiang Liu; Liwen Zhang. Optimized accelerated algorithms for large-scale dynamic low-rank matrix recovery. Under revision at JASA.
- (4) Bin Liu*; Yufeng Liu; Zhi Yang; Liwen Zhang. Fit First and Detect Later: A Unified Decoupled Framework for Change Detection in High Dimensions. Under review at JMLR.
- (5) Bin Liu*; Yufeng Liu; Zhi Yang; Liwen Zhang. Estimation of Multiple Structural Breaks in Ultra-High-Dimensional Matrix-Valued Graphical Models. Under review at JASA.
- (6) Qilong Ding; Bin Liu. Efficient multiple change point detection in time-varying Markov random fields. Under review at CSDA.
- (7) Bin Liu*; Yufeng Liu; Zhiqiang Liu; Liwen Zhang. Robust changepoint detection for low-rank matrix models under heavy-tailed noise. Under review at JASA.
- (8) Zhi Yang; Zhiqiang Liu; Liwen Zhang; Bin Liu. Split and Merge (SaM): Efficient Change-Point Detection with Additive Computational Cost in Low-Rank Matrices. Under review at Statistica Sinica.
- (9) Xianru Wang; Bin Liu*. A fast two-stage algorithm for multiple changepoint detection with general segment-wise losses. Under review at STAT.
- (10) Zhi Yang; Xianru Wang; Bin Liu; Yufeng Liu. Fast multiple change-point detection for dependent high-dimensional Gaussian graphical models. Under review at JOE.

Working Papers

- (1) Structured Change-Point Detection for Matrix Quantile Regression under Low-Rank and Sparse Constraints. Working paper.
- (2) Multiple Change-Point Detection in Structured Matrix Quantile Regression: A Fast Two-Step Procedure with Asymptotic Guarantees. Working paper.
- (3) Online Efficient Change Point Detection with Statistical Guarantees. Working paper.
- (4) Bayesian Inference for Change-Point Detection in High-Dimensional Matrix Graphical Models. Working paper.
- (5) Transfer Learning-Enhanced Change-Point Detection for Tensor Graphical Models. Working paper.
- (6) Diffusion-Assisted Change-Point Detection for High-Dimensional Data. Working paper.

RESEARCH GRANTS AND PROJECTS

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- 2016.01-2019.12 Testing correlation structures of high-dimensional random vectors. General Program, National Natural Science Foundation of China. Major participant.
 - 2018.01-2021.12 Statistical inference and applications for heterogeneous models in the big-data era. General Program, National Natural Science Foundation of China. Major participant.
 - 2020.01-2023.12 Statistical inference problems for high-dimensional big data. General Program, National Natural Science Foundation of China. Major participant.
 - 2022.01-2024.12 Statistical inference for parameter change points in high-dimensional linear regression models. Young Scientists Fund, National Natural Science Foundation of China. Principal investigator.
 - 2021.11-2022.06 Research on commodity price forecasting models. Shandong Zhuochuang Information Co., Ltd. Principal investigator.

- 2024.01-2028.12 Statistical inference for multisource heterogeneous data based on information fusion and transfer. Key Program, National Natural Science Foundation of China. Major participant.
- 2025.01-2028.12 Efficient statistical inference for change points in correlation structures of ultra-high-dimensional complex data. General Program, National Natural Science Foundation of China. Principal investigator.
- 2024.04-2025.12 Research on an intelligent business decision system for commodity data. Shandong Zhuochuang Information Co., Ltd. Principal investigator.

ACADEMIC SERVICE

Serves as an anonymous reviewer for domestic and international journals, including JASA, JRSSB, Statistica Sinica, Biometrics, AoAS, JBES, The American Statistician, JMVA, Computational Statistics, CSDA, Neurocomputing, Statistical Theory and Related Fields, Statistics and Computing, and Science China Mathematics.

PROFESSIONAL MEMBERSHIPS

- Council Member, Statistical Interdisciplinary Sciences Branch, Chinese Association for Applied Statistics, 2023.08.
- Member, China Statistical Education Society, 2022.10.
- Member, Digital Economy and Blockchain Technology Association, National Industrial Statistics Teaching and Research Association, 2020.10.
- Member, China Tourism Big Data Branch, Chinese Association for Applied Statistics, 2020.10.

CONFERENCE PRESENTATIONS AND INVITED TALKS

- East China Normal University, invited talk, Shanghai, April 21, 2026.
- 2026 Annual Meeting of the Young Statisticians Association, National Industrial Statistics Teaching and Research Association, invited talk, Chengdu, April 10-12, 2026.
- The 3rd National Joint Conference on Statistics and Data Science, contributed session talk, Hangzhou, July 11-14, 2025.
- Jiangsu Normal University, invited talk, online, July 2025.
- 2025 ICSA China Conference, contributed session talk, Zhuhai, June 27-30, 2025.
- The 13th National Conference on Probability and Statistics, invited talk, Xiamen, November 7-10, 2024.
- 2024 Bernoulli-IMS 11th World Congress in Probability and Statistics, contributed session talk, Germany, August 11-18, 2024.
- The 7th International Conference on Econometrics and Statistics, invited talk, Beijing, July 17-19, 2024.
- 2024 ICSA China Conference, organizer and invited talk, Wuhan, June 28-30, 2024.
- 2023 ICSA China Conference, organizer and invited talk, Chengdu, June 30-July 3, 2023.
- The 9th International Forum on Statistics, academic talk, Renmin University of China, July 14-15, 2023.
- The 1st National Joint Conference on Statistics and Data Science, Peking University, July 11-13, 2023.
- The 1st Conference on Machine Learning and Statistics, academic talk, East China Normal University, August 25-26, 2023.
- The 12th National Conference on Probability and Statistics, academic talk, Shandong University (Qingdao), April 13-16, 2023.

- 2022 International Symposium on Theory, Methods and Applications of Statistics and Data Science, Zhongnan University of Economics and Law, online, November 2022.
- The 2nd Forum for Outstanding Young Statisticians and Symposium on Digital Academic Research, Zhejiang Gongshang University, online, November 2022.
- Discipline Development Seminar of Statistics and Data Science, School of Management, Fudan University, academic talk, Shanghai, November 2021.
- Zhongtai Securities Institute for Financial Studies, Shandong University, online talk, October 2020.
- School of Economics, Xiamen University, online talk, April 2020.
- School of Statistics and Mathematics, Nanjing Audit University, academic talk, Nanjing, October 2019.

HONORS AND AWARDS

- First-Class Scholarship for Incoming Graduate Students, Fudan University, 2013.09.
- National Scholarship for Graduate Students, Fudan University, 2014.09.
- Third-Class Academic Scholarship for Ph.D. Students, Fudan University, 2016.09.
- First-Class Academic Scholarship for Ph.D. Students, Fudan University, 2018.09.
- Boxue Scholarship, Fudan University, 2018.09.
- Zheng Zukang Scholarship, Fudan University, 2018.10.
- First Prize for Conference Presentation, 2nd Shanghai University Statistics Ph.D. Student Forum, 2018.11.
- Oubei Cup Industrial Products Supply Chain Data Intelligence Innovation Competition, 2022.11.
- Excellence in Faculty and Staff Performance Assessment, School of Management, Fudan University, 2023.07.
- 2024 Young Faculty Star Award, School of Management, Fudan University, 2025.01.
- Second Prize, Shanghai Division, Contemporary Undergraduate Mathematical Contest in Modeling, 2025.12.

STUDENT SUPERVISION

* denotes Outstanding Thesis.

Undergraduate Students

- Class of 2026: Ji Di; Gao Youqing; Ouyang Jun; Du Zetao; Shen Yuqin.
- Class of 2025: Du Ying; Lu Haiyue*; Huang Rui*; Li Fangzheng.
- Class of 2024: Chi Zhenghao*; Wang Zerong; Wang Shuai; Zhou Lizhuo.
- Class of 2023: Qiu Ziyang; He Xuxuan*; Zhu Qingge.
- Class of 2022: Xu Tianzhang; Liu Wenyang; Zheng Zhi*; Ji Changhe.
- Class of 2021: Gong Hanqing; He Xinyi; Liu Yiting*; Wang Shengtao.

Master's Students

- Class of 2026, Master of Applied Statistics: Wu Siyuan*; Wang Jinglin; Tang Zining; Shi Wencheng*; Lan Zhengqi; Fang Keyi.
- Class of 2026, Master of Finance: Zhao Kangyang; Qian Jing; Chen Pengjun; Zhong Hanye*.

- Class of 2025, Master of Applied Statistics: Zhao Wenning; Xu Yiyuan*; Wang Ruotong; Man Hao*; Huang Sirui; Han Mingyue.
- Class of 2025, Master of Applied Statistics: Zhu Hongqiao; Yu Ruyi*; Liu Wenyang; He Yining; Chen Jiaqi*.

Ph.D. Students

- Class of 2024: Xianru Wang, co-supervised with Prof. Xinsheng Zhang; currently pre-tenure Associate Professor, School of Statistics and Data Science, Southwestern University of Finance and Economics.
- Class of 2025: Zeyu Li, co-supervised with Prof. Xinsheng Zhang; currently Postdoctoral Fellow, School of Statistics and Data Science, Southeast University.
- Class of 2026: Zhi Yang, co-supervised with Prof. Liwen Zhang at Shanghai University of Finance and Economics; currently Postdoctoral Fellow, Shanghai University of Finance and Economics.
- Current Ph.D. student: Zhiqiang Liu, co-supervised with Prof. Liwen Zhang at Shanghai University of Finance and Economics.